My Math Pocket Card

Y2 · 3rd term · test on 18 November

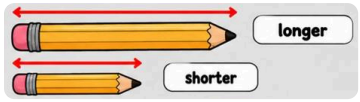
1 Measures · 2 Perimeter · 3 3D solids · 4 Data

FRONT

1 Words to compare and measure

Length is how long something is, from one end to the other. A pencil has a length.

long · short

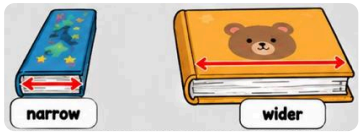


Long means how much something measures from one end to the other.

Short means not as long.

The top pencil is long. The bottom pencil is short.

wide · narrow

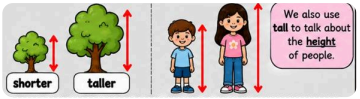


Wide means how much something measures across.

Narrow means not wide.

The big book is wide.

tall · height



Tall means how high something is, from the bottom to the top.

Height is how tall someone or something is. We also use tall for people.

The big tree is tall. The girl is tall.

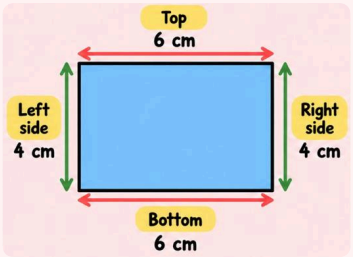
Compare two things: longer · shorter · wider · taller. ***The top pencil is longer than the bottom pencil.***

We measure length in **centimetres (cm)**, **metres (m)** and **kilometres (km)**.

2 Perimeter

distance around

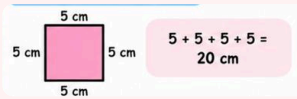
Perimeter is the total distance around a shape. **Around** means along all the sides of the shape. A **side** is one of the straight lines that make the outline of a flat shape.



Add all the sides!

Top	6 cm
Bottom	6 cm
Left side	4 cm
Right side	4 cm

6 + 6 + 4 + 4 = 20 cm

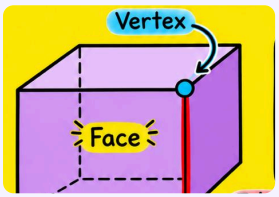


- Always write the number **and** the unit: **20 cm**.
- **Show your work**: write the sum, like 6 + 6 + 6 = 18 cm.

3 3D solids

faces · edges · vertices

A **3D solid** is a solid shape that has faces, edges and vertices.



face: a flat (straight) surface of a 3D solid. A cube has 6 faces.

edge: a straight line where two faces meet. A cube has 12 edges.

vertex: a corner where two or more edges meet. More than one vertex: **vertices**. A cube has 8 vertices.

Some solids, like the sphere, the cylinder and the cone, have a **curved surface**: a curved surface is round, not flat.



cube
6 faces
12 edges
8 vertices



cuboid
6 faces · 12 edges
8 vertices
like a cube, but the edges are not all the same length



pyramid
5 faces
8 edges
5 vertices



cylinder
2 flat faces
1 curved surface



cone
1 flat face
1 curved surface



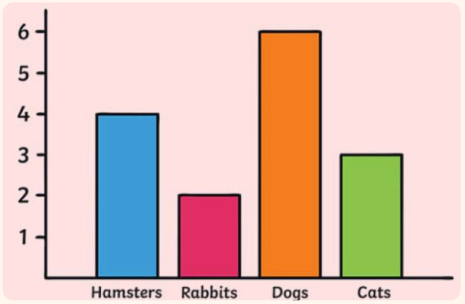
sphere
1 curved surface
no flat face, no edges, no vertices

Roll means turn over and over and move away, like a ball. A solid with a curved surface can roll: the sphere rolls. A cube has only flat faces: it cannot roll.

4 Data and bar graphs

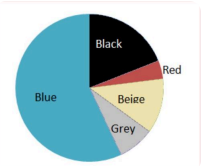
Data is all the information we collect and find out. We organize it using charts and graphs. A **piece of data** is one simple fact: ***My favorite color is blue.*** ***I have 2 pets.***

A **bar graph** shows data with bars. The taller the bar, the more there are. Look at the top of the bar and read the number on the side.



Hamsters	Rabbits	Dogs	Cats
4	2	6	3

- The most: **Dogs (6)**.
- The least: **Rabbits (2)**.
- How many cats? **3**.



The round picture is a **chart** too: it also shows data.

Make a bar graph: look at the table with numbers and colour each bar up to the right number.

5 Probability

likely · unlikely

Probability is thinking about if something will happen, won't happen, or will maybe happen.

certain	We know it will happen. ***I will eat dinner tonight.***
likely	Big chance to happen. ***You will find students in the school.***
possible	It will maybe happen. ***I will eat spaghetti for dinner.***
unlikely	Small chance to happen. ***You will see a penguin in the desert.***
impossible	We know it won't happen. ***I will eat rocks for dinner.***

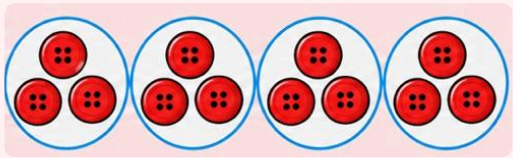
Say why with **because**: "It is impossible because ..."

6 Multiplication

making equal groups

Multiplication is a shorter way to show repeated addition. We use it to find the total when we have **equal groups**.

equal groups	have the same amount in each group.
repeated addition	is adding the same number again and again: 5 + 5 + 5.
multiplication sentence	is the math written with numbers and signs: 4 × 3 = 12.



3 + 3 + 3 + 3 = 12 → **4 × 3 = 12**

4 how many groups

3 how many in each group

4 groups of 3 buttons are 12 buttons altogether (in total).



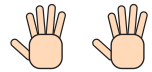
5 + 5 + 5 = 15 → **3 × 5 = 15**

3 groups of 5 flowers



2 + 2 + 2 + 2 + 2 = 10 → **5 × 2 = 10**

5 groups of 2 cookies



5 + 5 = 10 → **2 × 5 = 10**

2 hands, 5 fingers on each hand

7 Division

equal sharing

Division is a way to make equal groups of a number. We share a total into groups that have the same amount. **Share equally** means everybody gets the same amount. A **division sentence**: 10 ÷ 2 = 5.



10 ÷ 2 = 5

Mike and Laura share 10 oranges equally: 10 oranges shared into 2 equal groups is 5 oranges in each group.



8 ÷ 2 = 4

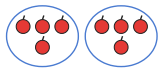
8 apples shared into 2 equal groups is 4 apples in each group.



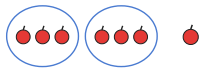
12 ÷ 3 = 4

12 cookies shared into 3 equal groups is 4 cookies in each group.

Even numbers (0, 2, 4, 6, 8, 10, 12...) can be shared equally into **2 groups**. **Odd numbers** (1, 3, 5, 7, 9...): when we share them between 2, **1 is left**.



8 between 2: 4 each



7 between 2: 3 each, 1 is left

How to write a good answer

key numbers

1. Read the question two times.
2. Start with words from the question: "*The cube has 6 faces.*"
3. Write the number **and** the unit: *18 cm*.
4. Show your work: write the math.
5. Say why with **because**.
6. Capital letter at the start, full stop at the end.

cube
6 · 12 · 8
faces · edges · vertices

pyramid
5 · 8 · 5
faces · edges · vertices

perimeter
add all the sides

certain › likely › possible › unlikely › impossible

4 × 3 = 12
12 ÷ 3 = 4

even: shared equally into 2 groups · odd: 1 is left

Math · Written answers

Year 2 · 3rd term · practice for the test

Name: _____ Date: _____

How to write a good answer

1. Read the question two times.
2. Start with words from the question: *"The cube has 6 faces."*
3. Write the number **and** the unit: *18 cm.*
4. Show your work: write the math.
5. Say why with **because**.
6. Capital letter at the start, full stop at the end.

1 · Length and perimeter

1 Look at the pencils. Which pencil is longer? Write a sentence.

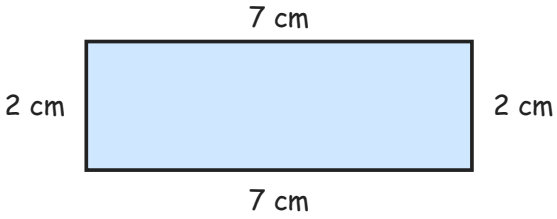
A 

B 

Pencil ____ is _____

_____

2 What is perimeter? Find the perimeter of the rectangle. Show your work.



Perimeter is

_____

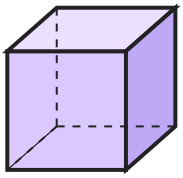
My work:

_____

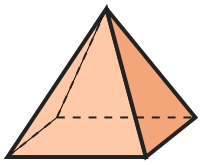
The perimeter is _____ cm.

2 · 3D solids

3 Count and complete the sentences.



cube

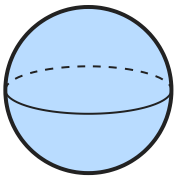


pyramid

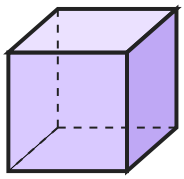
The cube has _____ faces, ___ edges and ___ vertices.

The pyramid has _____ faces, ___ edges and ___ vertices.

4 Which solid can roll? Write why. How many edges does the sphere have?



sphere



cube

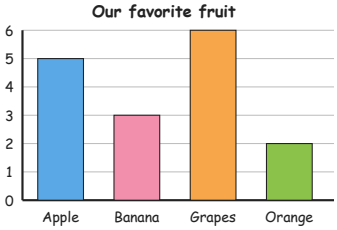
The _____ can roll because _____

_____

The sphere has _____ edges.

3 · Data and bar graphs

5 Look at the bar graph and answer with a sentence.



a) Which fruit do the most children like?

The most children like _____

b) Which fruit has the shortest bar?

_____ has the shortest bar. _____

c) How many children like bananas?

___ children like bananas. _____

4 · Probability

6 Write **certain**, **likely**, **possible**, **unlikely** or **impossible**.

certain · likely · possible · unlikely · impossible

a) You will eat dinner tonight. _____

b) You will find students in the school. _____

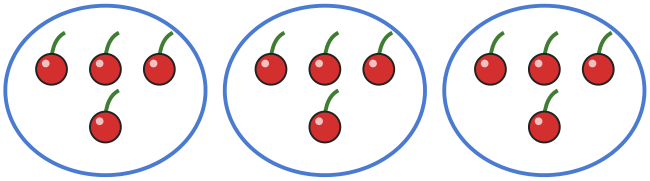
c) You will eat spaghetti for dinner. _____

d) You will see a penguin in the desert. _____

e) A cat will fly to the moon. _____

5 · Multiplication and division

7 How many cherries are in each group? Write the addition sentence. Write a multiplication sentence. What is multiplication?



In each group: _____

Addition sentence:

_____

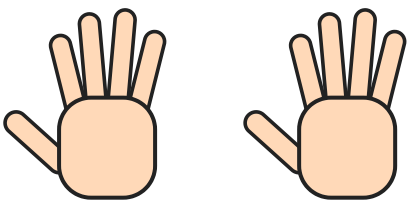
Multiplication sentence:

_____

Multiplication is

_____

8 How many fingers are on each hand? Write the addition sentence. Write a multiplication sentence.



On each hand: _____

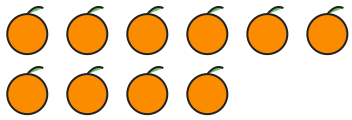
Addition sentence:

_____

Multiplication sentence:

_____

9 Mike and Laura share 10 oranges equally. How many oranges does each one get? Write the division sentence.



Division sentence:

_____

Each one gets _____ oranges.

10 Can 7 cookies be shared equally between 2 friends? Write why.



Yes / No: _____

Because

_____

_____

Great job! Check your answers with a grown-up. 🙂

Answers · check after you finish

1. Pencil A is longer.
2. The total distance around a shape. 7 + 7 + 2 + 2 = 18 cm.
3. Cube: 6, 12, 8. Pyramid: 5, 8, 5.
4. The sphere, because it has a curved surface. 0 edges.
5. a) Grapes. b) Orange has the shortest bar. c) 3
6. a) certain b) likely c) possible d) unlikely e) impossible
7. 4 + 4 + 4 = 12. 3 × 4 = 12. A shorter way to show repeated addition.
8. 5 + 5 = 10. 2 × 5 = 10.
9. 10 ÷ 2 = 5. Each one gets 5.
10. No. 7 is odd: 3 each and 1 is left.